

Probability practice

STAT UN 1001 · LECTURE 8

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Probability Thinking

Probability is about the chance of observing an event in a chance process.

Chance process generates one outcome at a time.

The outcome can be any of those in the sample space.

An event is defined *a description* of an outcome that belongs in the event.

According to this description, multiple outcomes (a subset) of the sample space correspond to an event.



Probability problem checklist

1. Chance process
2. Sample space
3. Event of interest
4. Probability sentence
5. Rule or strategy

We will write the sentence before computing the number.

Joint, marginal, conditional

Question type	Plain-language cue	Example probability
Joint	Both things happen	P(A and B)
Marginal	One thing happens, ignoring the other variable	P(A)
Conditional	One thing happens among a restricted group	P(A | B)

READING TIP Circle words like “and,” “or,” “given,” “among,” and “at least.” They often tell you the probability structure.

The hardest part is often naming the denominator.

Probability tree logic

BRANCH RULE

Branches from the same split add to 1.

PATH RULE

Multiply along a path to get a joint probability.

EVENT RULE

Add the path probabilities that satisfy the event.

Start
A, P(A)
B, P(B | A)
C, P(C | A and B)

Path probability:
P(A and B and C)
= P(A) x P(B | A) x P(C | A and B)

Practice 1: Pokemon pack

A promotion says that each pack independently has a 1 in 20 chance of containing a rare card. A student buys 3 packs.

What is the probability that at least one of the 3 packs contains a rare card?

First: describe the chance process in ordinary language.

Pokemon pack setup

CHANCE PROCESS

Open 3 independent packs. Each pack can be Rare or Not rare.

SAMPLE SPACE

RRR, RRN, RNR, NRR, RNN, NRN, NNR, NNN

EVENT

At least one rare card: all outcomes except NNN.

PROBABILITY SENTENCE

P(at least one rare in 3 packs) = 1 - P(no rare in 3 packs)

Rule: complement. Answer: 1 - (19/20)^3 = 0.1426, about 14%.

Practice 2: Cognitive screening

In a campus pilot study, 120 volunteers take a short attention task and answer a sleep question.

	Low sleep	Typical sleep	Total
Flagged by task	18	12	30
Not flagged	22	68	90
Total	40	80	120

Find the marginal probability of low sleep, the joint probability of low sleep and flagged, and P(low sleep | flagged).

First: which probability uses all 120 volunteers as the denominator?

Cognitive screening setup

CHANCE PROCESS

Randomly select one volunteer from this pilot study.

SAMPLE SPACE

The 120 volunteers, organized by task result and sleep report.

PROBABILITY SENTENCES

P(low sleep) = 40 / 120
P(low sleep and flagged) = 18 / 120
P(low sleep | flagged) = 18 / 30

Marginal: 0.333. Joint: 0.150. Conditional: 0.600.

Cognitive screening tree

One useful tree starts with task result, then splits by sleep report.

Volunteer
- Flagged, 30/120 = 0.25
  - Low sleep, 18/30 = 0.60
  - Typical sleep, 12/30 = 0.40
- Not flagged, 90/120 = 0.75
  - Low sleep, 22/90 = 0.244
  - Typical sleep, 68/90 = 0.756

JOINT FROM ONE PATH

P(flagged and low sleep)
= P(flagged) x
P(low sleep | flagged)

0.25 x 0.60 = 0.15

Practice 3: Quality check

A company tests 1,000 devices before shipping. About 3% have a hidden defect. The test flags 90% of defective devices and falsely flags 8% of non-defective devices.

If a device gets flagged, what is the probability that it truly has the defect?

HINT

Use a table with expected counts out of 1,000 devices.

Quality check setup

	Test Outcome		
True status	Flagged	Not flagged	Total
Defective	27	3	30
Not defective	78	892	970
Total	105	895	1,000

PROBABILITY SENTENCE

P(defective | flagged) = count(defective and flagged) / count(flagged)

RULE

Conditional probability. Natural frequencies make the denominator visible.

Answer: 27 / 105 = 0.257, about 26%.